

Electricity Bill Generator

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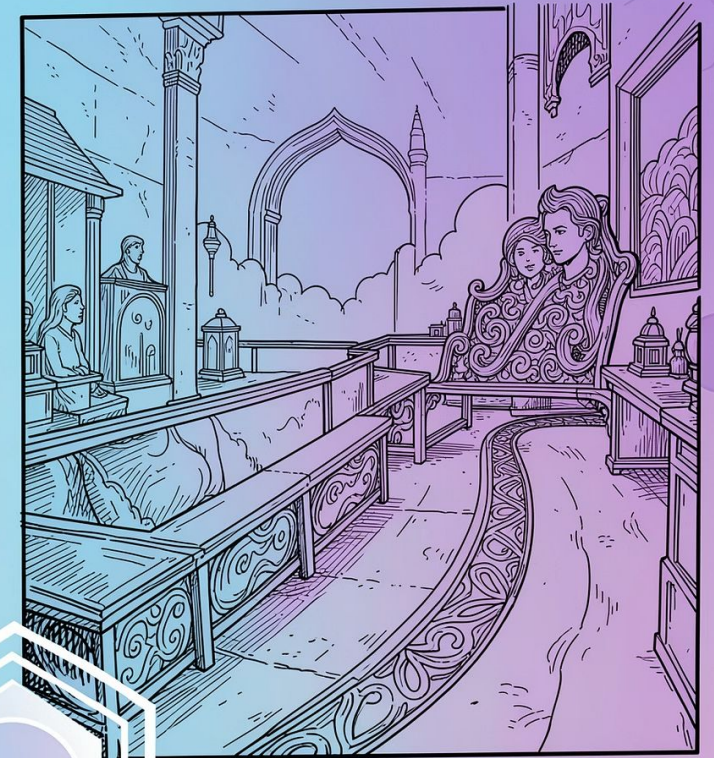
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Agenda

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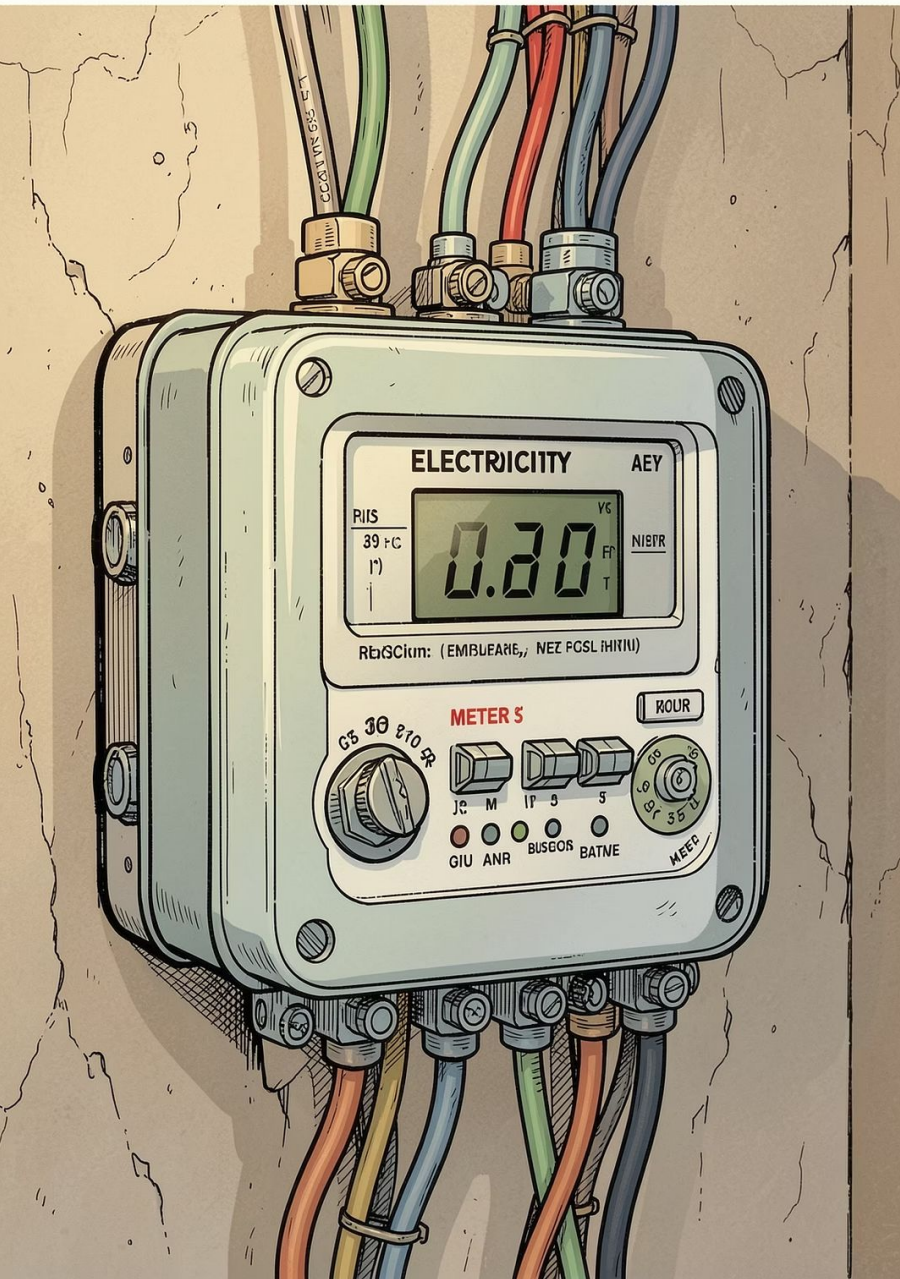
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Introduction

Electricity is essential in every home. This project automatically computes monthly electricity bills using C functions and if-else conditional logic to reduce manual errors and deliver fast, accurate results.

Problem Statement

Write a C program that calculates the electricity bill using slab rates:

0 – 100 units

₹1.5 per unit

101 – 200 units

₹2.5 per unit

Above 200 units

₹4 per unit

❏ Maintain exact slab logic to ensure billing accuracy.

Objectives

- **Objective 01**

Automatically calculate electricity bills.

- **Objective 02**

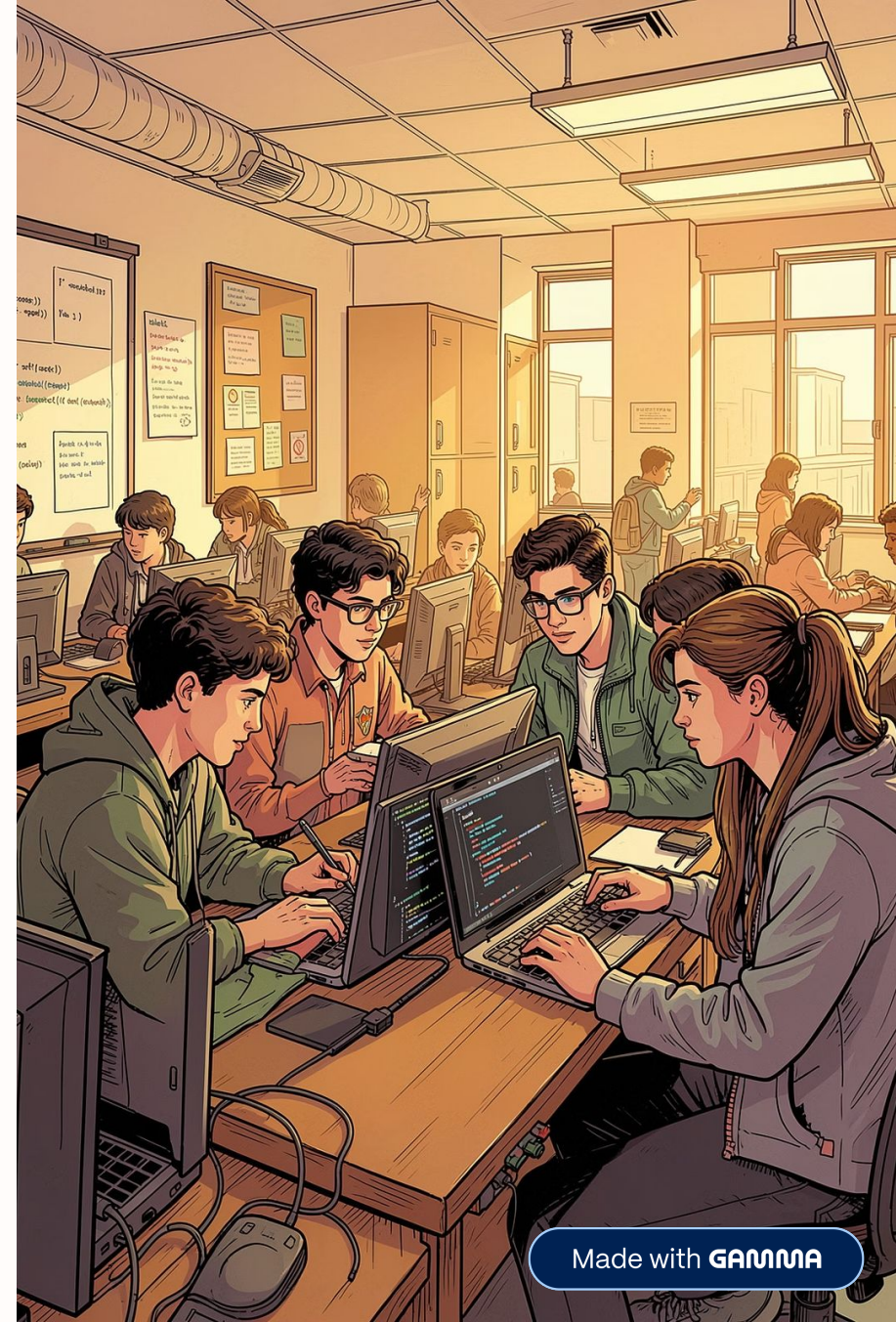
Understand if-else conditional statements in C.

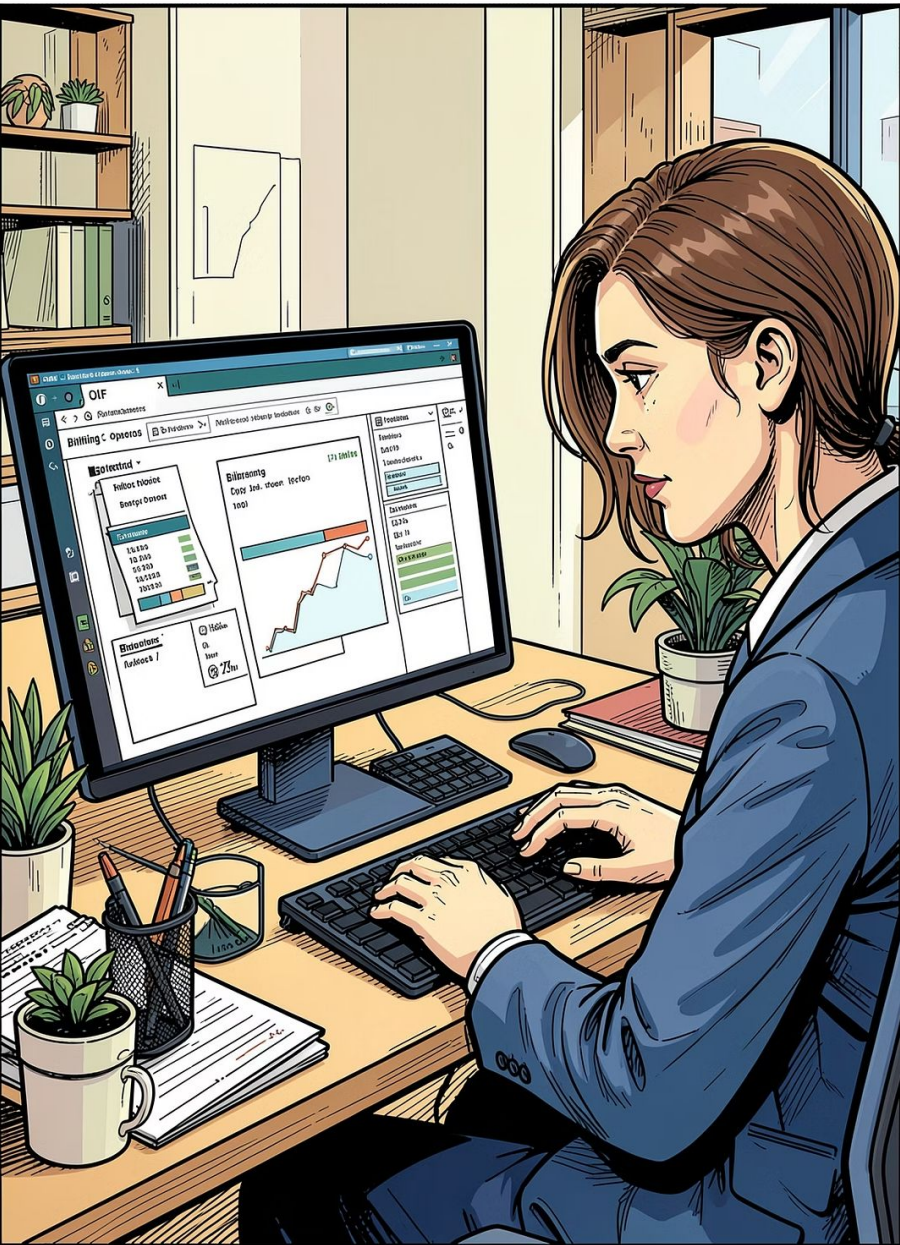
- **Objective 03**

Learn how functions improve code modularity.

- **Objective 04**

Reduce manual calculation errors.





User Stories

Household Consumer

Enter units to quickly get the monthly bill without manual math.

Billing Operator

Automatically apply correct slab rates for fast, accurate bills.

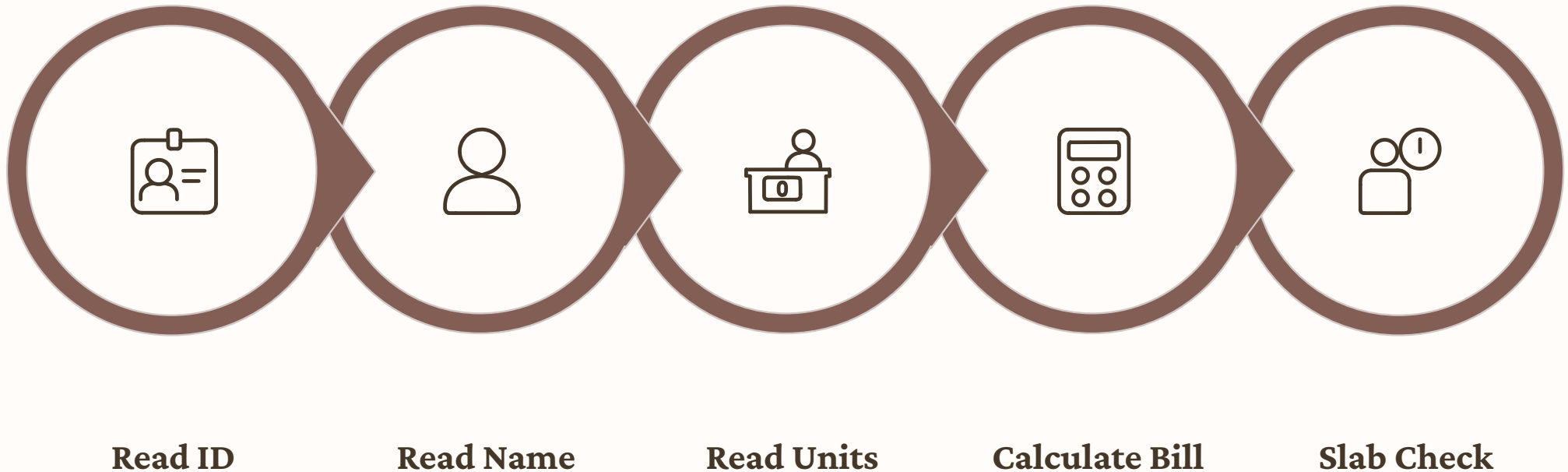
Consumer Monitoring

Calculate bills for different unit values to understand cost impact.

Electricity Department

Automate billing to save time and avoid calculation mistakes.

Program Logic (Flow)



This straightforward flow ensures clear separation between input, computation, and output.

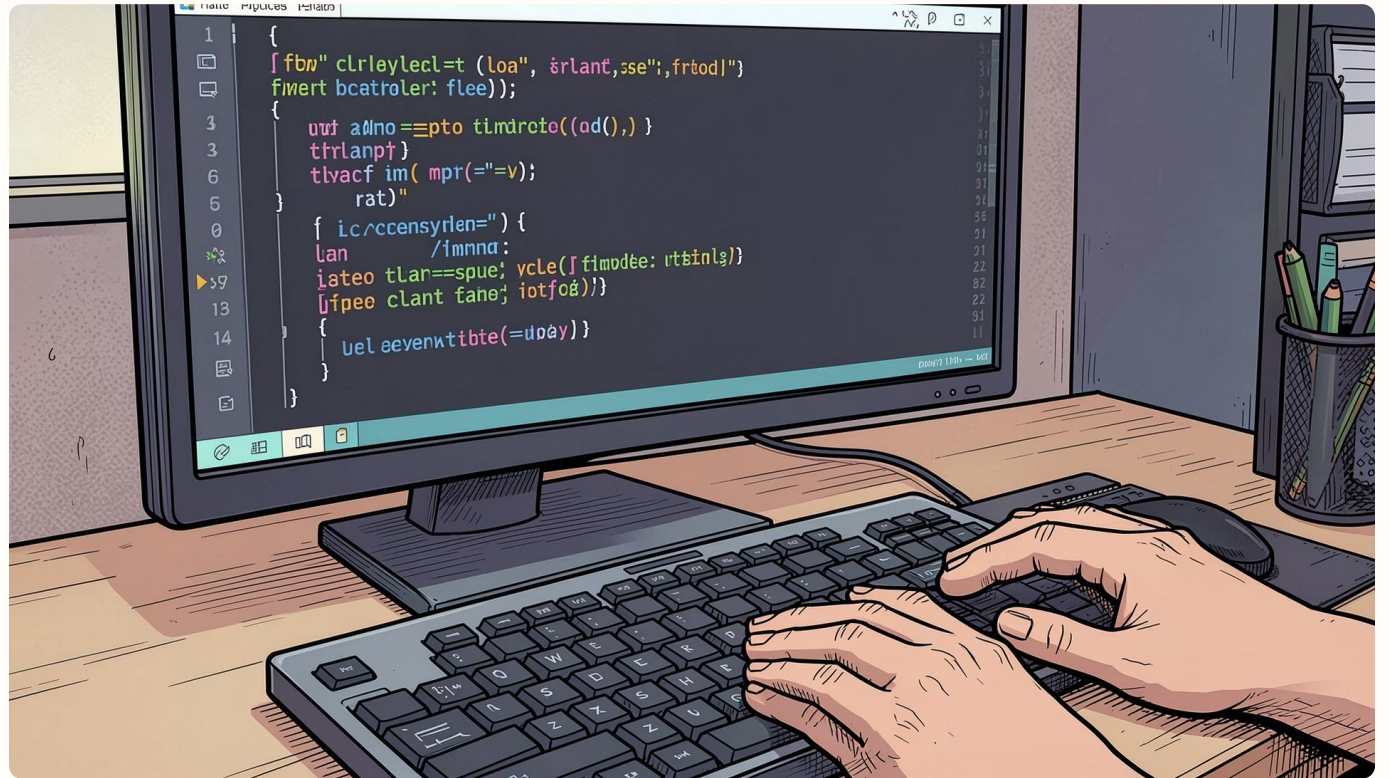
Algorithm & Core Function

Algorithm summary:

- Start → Read customer ID, name, units
- Call calculateBill(units) → apply slabs
- Return and print bill → Stop

Core calculation:

```
float calculateBill(int units) {  
    if(units<=100) bill=units*1.5; else  
    if(units<=200) bill=(100*1.5)+((units-  
    100)*2.5); else bill=(100*1.5)+  
    (100*2.5)+((units-200)*4); return bill;  
}
```



Sample Outputs (Verified)

```
Enter Customer ID: 108474804
Enter Customer Name: AMISHA
Enter Electricity Units Consumed: 75
```

```
=====
ELECTRICITY BILL
=====
```

```
Customer ID   : 108474804
Customer Name : AMISHA
Units Consumed: 75
Total Bill    : Rs. 112.50
=====
```

```
PS C:\Users\khush\Desktop\project21> █
```

```
Enter Customer ID: 100959214
Enter Customer Name: Aashi
Enter Electricity Units Consumed: 150
```

```
=====
ELECTRICITY BILL
=====
```

```
Customer ID   : 100959214
Customer Name : Aashi
Units Consumed: 150
Total Bill    : Rs. 275.00
=====
```

```
PS C:\Users\khush\Desktop\project21> █
```

```
Enter Customer ID: 1074809210
Enter Customer Name: BhumiKa
Enter Electricity Units Consumed: 250
```

```
=====
ELECTRICITY BILL
=====
```

```
Customer ID   : 1074809210
Customer Name : BhumiKa
Units Consumed: 250
Total Bill    : Rs. 600.00
=====
```

```
PS C:\Users\khush\Desktop\project21> █
```

Examples from runs:

- Units 75 → Total Bill: Rs. 112.50
- Units 150 → Total Bill: Rs. 275.00
- Units 250 → Total Bill: Rs. 600.00

Future Improvements & Conclusion



Save Records

Persist bills to a file or database for history and reporting.



GST & Fixed Charges

Include taxes and fixed fees for realistic bills.



Menu-Driven Program

Allow multiple operations: add, view, update, delete records.



Graphical Interface

Build a user-friendly GUI for non-technical users.

Conclusion: This project demonstrates modular programming in C—functions + if-else logic—to solve a practical billing problem reliably. It reduces manual errors and is ready for enhancements (GST, persistence, GUI).

☐ Thank you — questions welcome.

